Here is the engine of the proof. By Prop. I the areal velocity is constant, so the periodic time is the whole area πab divided by that steady rate. By Cor. Prop. XIV the lesser axis b is the mean proportional between the greater axis a and the latus rectum L, giving $b^2 = aL$, whence $b \propto \sqrt{aL}$. Since the force is as $1/SP^2$ (Prop. XI), the latus rectum enters only in its subduplicate ratio; subducting that half-power leaves the sesquiplicate ratio $T \propto a^{3/2}$, that is $T^2 \propto a^3$. Thus from ellipse, area, and inverse-square, Kepler's crown descends by pure geometry. *Q.E.I.*